lies on features in the impressions made by distinct ridges on fingertips

4. Face recognition that identifies individuals by analysing facial features such as the upper outlines of eye sockets or sides of the mouth

5. Voice recognition that is based on the differences in voice

Data mining and analysis technologies. Data mining and analysis technologies are used for analysing historical and current online data for prediction and description by identifying patterns and anomalies from the vast database.

Prediction involves using variables in the database to predict unknown or future values of other variables of interest.

Description involves increasing knowledge about a variable or dataset by finding related information to obtain human-interpretable patterns describing the data.

The trail of terrorists is often indistinguishable from mass commercial and government data. Traditional investigation techniques follow an intricate process of seeking out information related to an individual. This becomes increasingly problematic as more and more information becomes available.

Network-centric operations. These operations increase the operational effectiveness and efficiency by networking sensors, decision makers, law-enforcement agencies and disaster managers to achieve increased speed of operations, increased security and safety, reduced vulnerability to hostile action and self-synchronisation. Systems from local to national levels can be integrated into a network to address antiterrorism.

Fig. 5: Telegram is an app widely used by ISIS since it offers double encryption (Image courtesy: www.bidnessetc.com)

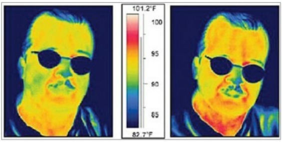

Fig. 6: Facial thermographs, a biometric technology likely to become popular in the near future (Image courtesy: http://biometrics.mainguet.org)

How terrorist groups are using IT

Terrorist groups all over the world are widely using IT for various functions such as internal communication and coordination, propaganda and misinformation, recruitment and financing, information and intelligence gathering with the final objective of spreading hate and violence.

Islamist terrorist group *Al-Qaida* created by Osama Bin Laden relies heavily on computers and other modern communication tools. By the early 2000s, Islamist terrorists had started using chat rooms, pornographic sites and other Internet utilities to masquerade maps and photographs of their targets, and for sharing directions and information for their operations.

US officials stumbled upon messages that were encoded using free encryption programs set up by Internet privacy groups. Images were formed via a series of dots inside which were strings of letters and numbers that computers could read to reconstruct images. These are just a few examples of how IT has been used by terror groups.

Irish terrorists took help of contract hackers to penetrate computers to get the home addresses of law-enforcement and intelligence officers. In March 2000, an investigation by Japanese police forces showed that a software system had been bought by Aum Shinrikyo cult, which is notorious for poisoning Tokyo subway in 1995. This software system was used to monitor more than 100 police vehicles, and was also sold to several Japanese companies and government agencies, thus, making them vulnerable to cyber attacks by this cult group.

Today, terrorists from the infamous ISIS group meet openly on Facebook or Twitter. They ensure their discussions remain highly confidential by using encryption technology to encode their messages. ISIS has a technology cell that consists of five to six members offering 24/7 support on encrypting communications, disguising personal details and using apps like Twitter while avoiding surveillance.

ISIS is known to use an app called Telegram, which offers two layers of encryption, and claims to be faster and more secure than the more commonly-used WhatsApp.

The issue of hunting out terrorist communication is almost like finding a needle in a haystack. It is not possible to scan every voice call, message or email, given the very high volume of communication. Then, there are problems related to the right to privacy and confidentially of certain communication such as business plans and banking transactions.

Safeguard technologies for the future

Current levels of technology available are actually inadequate to deal with the scope and potential severity of the terrorism threat. You need to always remain one step ahead of the terrorists and develop security systems that are capable of protecting the public,